

FMUSER FUQ-4415EF Transmodulator 4 Tuner Inputs DVB-S-S2-S2X DVB-C-T-T2-ISDB-T Input 4 DVB-C RF 2 DVB-T RF Outputs BISS Descrambling PID Pass-Through Broadcast Cable TV Redistribution

Product overview



The FMUSER FUQ-4415EF is a transmodulator with 4 tuner inputs for DVB-S/S2/S2X or DVB-C/T/T2/ISDB-T standards, and outputs 4 DVB-C RF or 2 DVB-T RF signals. It demodulates, re-multiplexes, and modulates signals with embedded BISS descrambling for secure processing. This unit features LCD/keyboard control and web-based NMS for operations and upgrades. Ideal for broadcast, cable TV distribution, and secure content redistribution.

Key Features

Integrated demodulating, re-multiplexing, BISS descrambling, and modulating in one unit.

Supports 4 tuner inputs for DVB-S/S2/S2X or DVB-C/T/T2/ISDB-T standards.

Provides 4 DVB-C RF or 2 DVB-T RF outputs for versatile distribution.

Enables PID pass-through for seamless program identifier transmission.

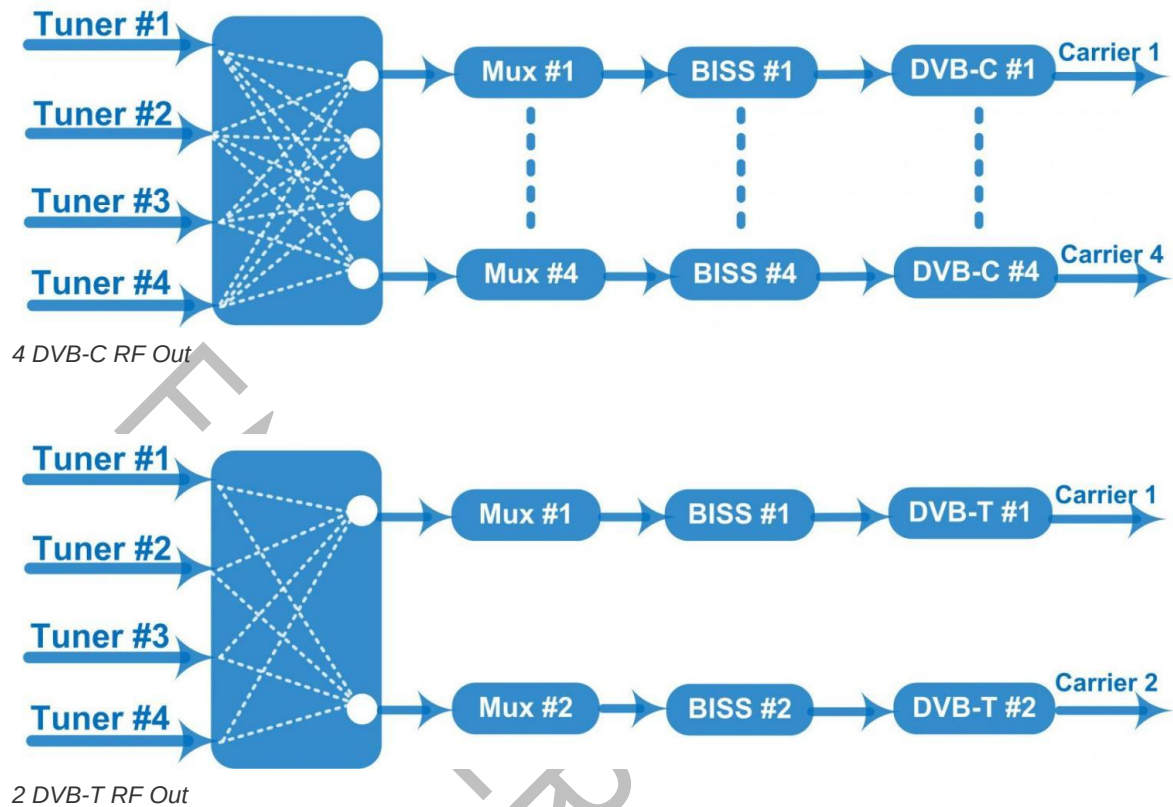
Features LCD/keyboard and web-based NMS for control and upgrades. Supports 4 tuner inputs for DVB-S/S2/S2X or DVB-C/T/T2/ISDB-T standards.

Provides 4 DVB-C RF or 2 DVB-T RF outputs for versatile distribution.

Enables PID pass-through for seamless program identifier transmission.

Features LCD/keyboard and web-based NMS for control and upgrades.

Principle Chart



Specifications

Input	Input	4 Tuner input (3*D VB-S/S2/ S2X+1*D VB-S/S2/ S2X or 1* DVB-C/T/ T2/ISDB-T)	4 Tuner input (3*D VB-S/S2/ S2X+1*D VB-S/S2/ S2X or 1* DVB-C/T/ T2/ISDB-T)	4 Tuner input (3*D VB-S/S2/ S2X+1*D VB-S/S2/ S2X or 1* DVB-C/T/ T2/ISDB-T)	4 Tuner input (3*D VB-S/S2/ S2X+1*D VB-S/S2/ S2X or 1* DVB-C/T/ T2/ISDB-T)	4 Tuner input (3*D VB-S/S2/ S2X+1*D VB-S/S2/ S2X or 1* DVB-C/T/ T2/ISDB-T)	4 Tuner input (3*D VB-S/S2/ S2X+1*D VB-S/S2/ S2X or 1* DVB-C/T/ T2/ISDB-T)
Tuner Section	Multi-mode tuners switchable	DVB-C	Input Frequency	Input Frequency	Input Frequency	60 MHz~890 MHz	60 MHz~890 MHz
Tuner Section	Multi-mode tuners switchable	DVB-C	Standard	Standard	Standard	J.83A (DVB-C), J.83C	J.83B
Tuner Section	Multi-mode tuners switchable	DVB-C	Constellation	Constellation	Constellation	16/32/64/128/256 QAM	64/256 QAM
Tuner Section	Multi-mode tuners switchable	DVB-T/T2	Input Frequency	Input Frequency	Input Frequency	60 MHz~890 MHz	60 MHz~890 MHz

Tuner Section	Multi-mode tuners switchable	DVB-T/T2	Bandwidth	Bandwidth	Bandwidth	5/6/7/8 M bandwidth	5/6/7/8 M bandwidth
Tuner Section	Multi-mode tuners switchable	ISDB-T	Input Frequency	Input Frequency	Input Frequency	60 MHz~890 MHz	60 MHz~890 MHz
Tuner Section		DVB-S	Input Frequency	Input Frequency	Input Frequency	950-2150 MHz	950-2150 MHz
Tuner Section		DVB-S	Symbol rate	Symbol rate	Symbol rate	QPSK: 0.5~45 Msps	QPSK: 0.5~45 Msps
Tuner Section		DVB-S	FEC Demodulation	FEC Demodulation	FEC Demodulation	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Tuner Section		DVB-S	Max input bitrate	Max input bitrate	Max input bitrate	≤140 Mbps	≤140 Mbps
Tuner Section		DVB-S2	Input Frequency	Input Frequency	Input Frequency	950-2150 MHz	950-2150 MHz
Tuner Section		DVB-S2	Symbol rate	Symbol rate	Symbol rate	QPSK/8PSK/16APSK: 0.5~45 Msps; 32APSK: 0.5~40 Msps	QPSK/8PSK/16APSK: 0.5~45 Msps; 32APSK: 0.5~40 Msps
Tuner Section		DVB-S2	FEC Demodulation	FEC Demodulation	FEC Demodulation	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10

Tuner Section		DVB-S2	Max input bitrate	Max input bitrate	Max input bitrate	≤140 Mbps	≤140 Mbps
Tuner Section		DVB-S2X	Input Frequency	Input Frequency	Input Frequency	950-2150 MHz	950-2150 MHz
Tuner Section		DVB-S2X	Symbol rate	Symbol rate	Symbol rate	QPSK/8PSK/16APSK: 0.5~45 Msps; 8APSK: 0.5~40 Msps; 32APSK: 0.5~40 Msps	QPSK/8PSK/16APSK: 0.5~45 Msps; 8APSK: 0.5~40 Msps; 32APSK: 0.5~40 Msps

Tuner Section		DVB-S2X	FEC Demodulation	FEC Demodulation	FEC Demodulation	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10, 2/3-L, 32/45, 11/15, 7/9	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10, 2/3-L, 32/45, 11/15, 7/9
Tuner Section		DVB-S2X	Max input bitrate	Max input bitrate	Max input bitrate	≤140 Mbps	≤140 Mbps
Multiplexing	Multiplexing	Maximum PID Remapping	128 per input channel	128 per input channel	128 per input channel	128 per input channel	128 per input channel

Multiplexing	Multiplexing	Function	PCR accurate adjusting	PCR accurate adjusting	PCR accurate adjusting	PCR accurate adjusting	PCR accurate adjusting
Multiplexing	Multiplexing	Function	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically
Multiplexing	Multiplexing	Function	PID Remapping	PID Remapping	PID Remapping	PID Remapping	PID Remapping
BISS Descrambling	BISS Descrambling	Mode 1, Mode E (Up to 60Mbps) (descramble individual program)	Mode 1, Mode E (Up to 60Mbps) (descramble individual program)	Mode 1, Mode E (Up to 60Mbps) (descramble individual program)	Mode 1, Mode E (Up to 60Mbps) (descramble individual program)	Mode 1, Mode E (Up to 60Mbps) (descramble individual program)	Mode 1, Mode E (Up to 60Mbps) (descramble individual program)
Modulation	Modulation	DVB-C	Standard	Standard	J.83A (DVB-C), J.83B, J.83BC	J.83A (DVB-C), J.83B, J.83BC	J.83A (DVB-C), J.83B, J.83BC
Modulation	Modulation	DVB-C	Constellation	Constellation	J.83A	J.83B	J.83C
Modulation	Modulation	DVB-C	Constellation	Constellation	16/32/64/128/256QAM	64/256 QAM	64/256 QAM
Modulation	Modulation	DVB-C	Bandwidth	Bandwidth	8M	6M	6M
Modulation	Modulation	DVB-C	MER	MER	≥43dB	≥43dB	≥43dB
Modulation	Modulation	DVB-C	Symbol rate	Symbol rate	5.000~7.000Msps adjustable	5.000~7.000Msps adjustable	5.000~7.000Msps adjustable
Modulation	Modulation	DVB-C	RF Out	RF Out	F typed, 75Ω impedance	F typed, 75Ω impedance	F typed, 75Ω impedance
Modulation	Modulation	DVB-C	RF Out	RF Out	4*DVB-C RF output (adjacent carriers)	4*DVB-C RF output (adjacent carriers)	4*DVB-C RF output (adjacent carriers)

Modulation	Modulation	DVB-C	RF frequency	RF frequency	50~960M Hz, 1KHz step	50~960M Hz, 1KHz step	50~960M Hz, 1KHz step
Modulation	Modulation	DVB-C	RF output level	RF output level	-20~-5dBm, 1 dBm step	-20~-5dBm, 1 dBm step	-20~-5dBm, 1 dBm step
Modulation	Modulation	DVB-C	Output Bitrate	Output Bitrate	Max 51.6 Mbps per channel	Max 51.6 Mbps per channel	Max 51.6 Mbps per channel
Modulation	Modulation	DVB-T	Standard	Standard	EN300744	EN300744	EN300744
Modulation	Modulation	DVB-T	Constellation	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulation	Modulation	DVB-T	Bandwidth	Bandwidth	6M, 7M, 8M	6M, 7M, 8M	6M, 7M, 8M
Modulation	Modulation	DVB-T	MER	MER	≥42 dB	≥42 dB	≥42 dB
Modulation	Modulation	DVB-T	Guard Interval	Guard Interval	1/4, 1/8, 1/16, 1/32	1/4, 1/8, 1/16, 1/32	1/4, 1/8, 1/16, 1/32
Modulation	Modulation	DVB-T	FFT mode	FFT mode	2K, 4K, 8K	2K, 4K, 8K	2K, 4K, 8K
Modulation	Modulation	DVB-T	MER	MER	≥42 dB	≥42 dB	≥42 dB
Modulation	Modulation	DVB-T	Code Rate	Code Rate	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Modulation	Modulation	DVB-T	RF out	RF out	F typed, 75Ω impedance	F typed, 75Ω impedance	F typed, 75Ω impedance
Modulation	Modulation	DVB-T	RF out	RF out	2*DVB-T RF output	2*DVB-T RF output	2*DVB-T RF output
Modulation	Modulation	DVB-T	RF frequency	RF frequency	30~999.999MHz, 1KHz step	30~999.999MHz, 1KHz step	30~999.999MHz, 1KHz step

Modulation	Modulation	DVB-T	RF output Level	RF output Level	-20~-5 dBm, 1 dBm step	-20~-5 dBm, 1 dBm step	-20~-5 dBm, 1 dBm step
Modulation	Modulation	DVB-T	Output Bitrate	Output Bitrate	Max 31.6 68Mbps per channel	Max 31.6 68Mbps per channel	Max 31.6 68Mbps per channel
System	System	LCD/keyboard and web management	LCD/keyboard and web management	LCD/keyboard and web management	LCD/keyboard and web management	LCD/keyboard and web management	LCD/keyboard and web management
System	System	Language: English	Language: English	Language: English	Language: English	Language: English	Language: English
System	System	upgrade via Web	upgrade via Web	upgrade via Web	upgrade via Web	upgrade via Web	upgrade via Web
General	General	Dimension (W×L×H)	Dimension (W×L×H)	482mm×180mm×44mm	482mm×180mm×44mm	482mm×180mm×44mm	482mm×180mm×44mm
General	General	Approx weight	Approx weight	4 kg	4 kg	4 kg	4 kg
General	General	Environment	Environment	0~45■(Work)■-20~-80■(Storage)■	0~45■(Work)■-20~-80■(Storage)■	0~45■(Work)■-20~-80■(Storage)■	0~45■(Work)■-20~-80■(Storage)■
General	General	Power requirements	Power requirements	100~240 VAC, 50/60Hz	100~240 VAC, 50/60Hz	100~240 VAC, 50/60Hz	100~240 VAC, 50/60Hz

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

Contact:Tomleequan Email■sales@fmuser.com whatsapp: [+8613922702227](https://wa.me/8613922702227)